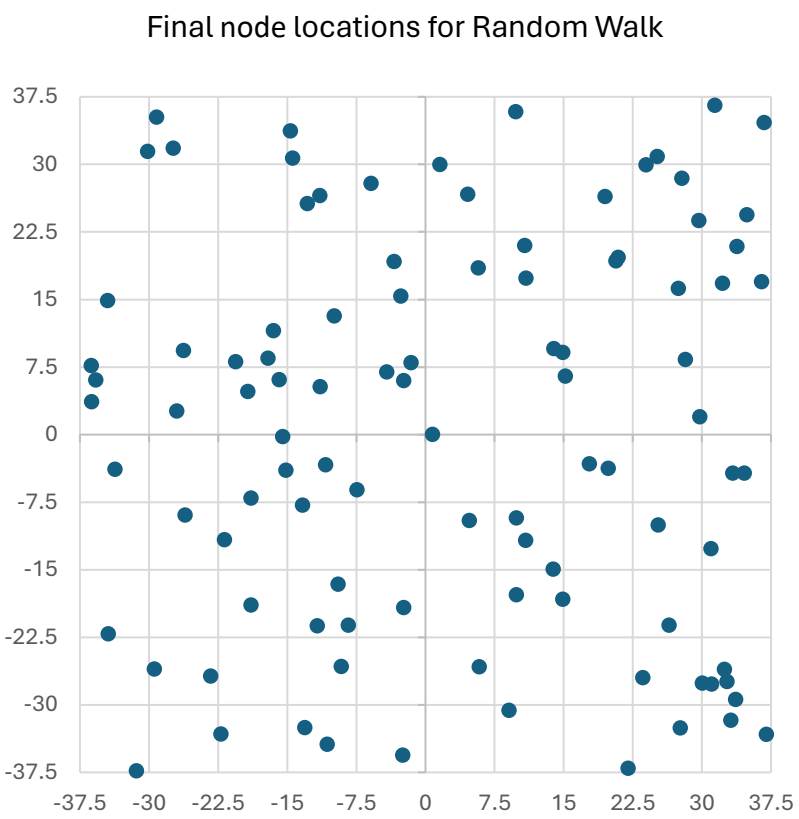
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ECE 6110 Lab Assignment 3
Mobility Models

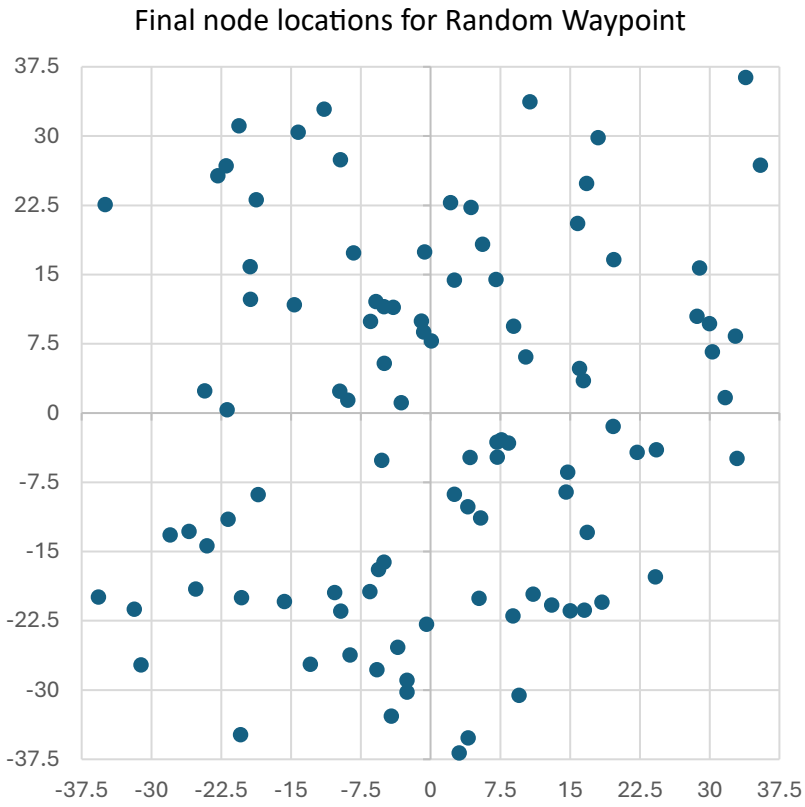
Part 1: Comparison of Mobility Models

a) In this lab, I implemented first a program using ns3 that simulates the movement of 100 nodes in a 75m x 75m square area using three different mobility models: random walk, random waypoint, and random direction. We analyzed the output of the program in terms of the final positions of the nodes and their average distance from the center point of the region.

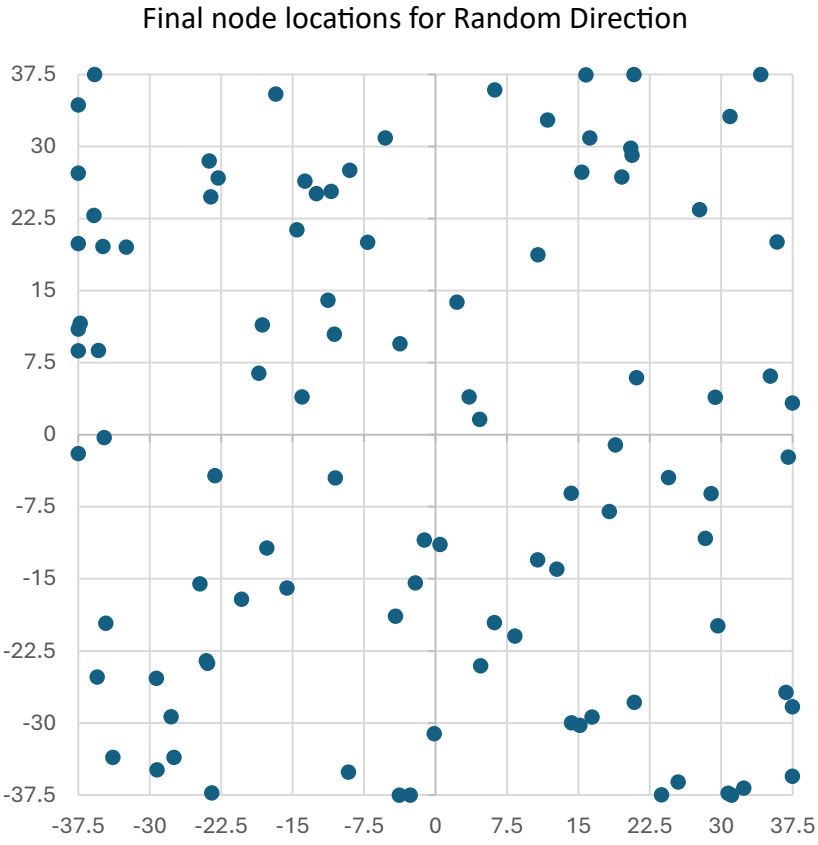
The following figures show the plots of the locations of the 100 nodes at the end of the simulation time for each mobility model. Their speed was between 2.0 and 6.0 $m.s^{-1}$, with a pause time of 2.0 s and a simulation duration of 400 s.



For Random Walk, the average distance from the center point is: 28.9700 m.



For Random Walk, the average distance from the center point is: 23.7100 m.



For Random Walk, the average distance from the center point is: 32.1903 m.

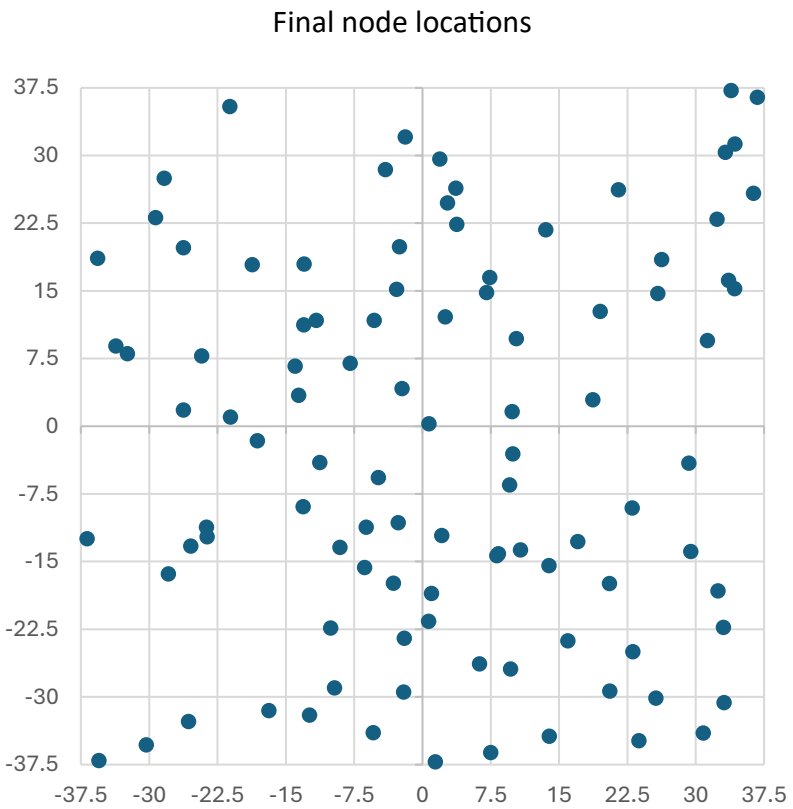
The results show that the random direction mobility model produces the highest average distance from the center point of the region, followed by the random walk mobility model, and then the random waypoint mobility model.

With the random direction mobility model each node moves with a random speed and direction and changes its direction randomly after a random time. This means that the nodes may move a little bit closer to the perimeter and the corners of the region compared to the other algorithm. Therefore, the average distance from the center point of the region is the highest.

For the random walk mobility model, each node moves with a constant speed and changes its direction randomly after a fixed distance step. This means that the nodes tend to stay close to their initial positions, and do not explore the whole region. Therefore, the average distance from the center point of the region is relatively low compared to random direction.

Using the random waypoint mobility model, each node moves with a random speed and direction towards a random destination and pauses for a certain time before choosing a new destination. This means that the nodes tend to cover more area and have a higher chance of reaching the edges of the region. However, the pause time also reduces the mobility of the nodes and makes them stay at the same location for a while, which explains why the average distance is lower than for the random walk.

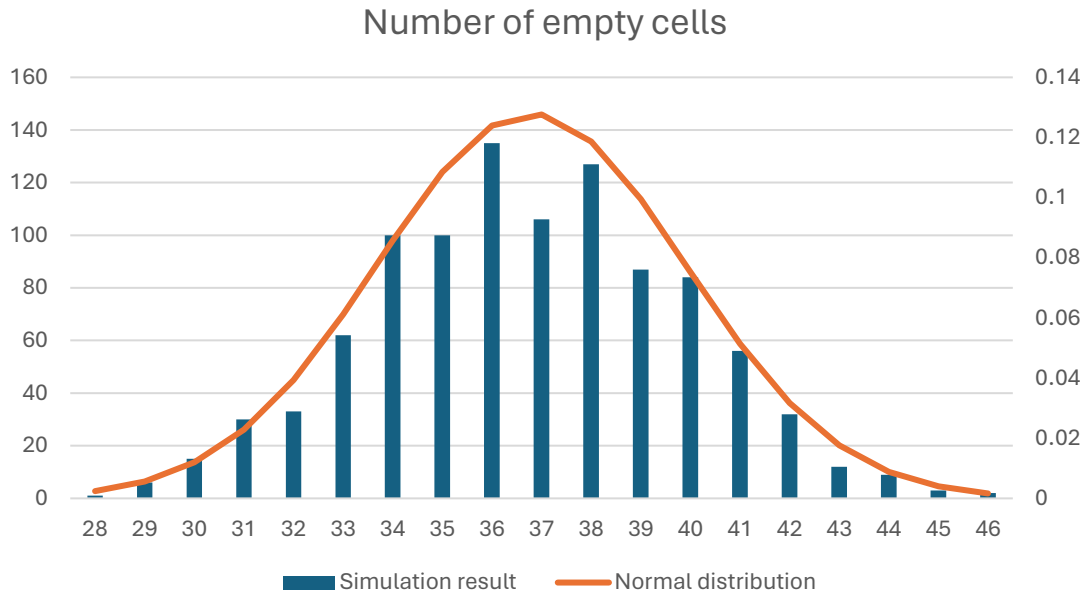
b) We now redo the same simulation for the random waypoint model, but with a speed between 20 and 22 $m.s^{-1}$ and a pause time of 60 s.



This time, the average distance from the center point is: 27.2129 m.

This time, compared to the result for the random waypoint in part 1a, we observe that the average node distance from the center point is higher. It can be explained by the fact that the nodes have a higher speed and a longer pause time, which makes them more likely to get close to the border of the area and stay there. The final node location is also a little bit more uniform and closer to the one of the random walk.

We now apply the number of empty cell test to this configuration. To this end, we must divide the area into $n = 100$ cells. Then, we can place each node in the cell in which it ended the simulation. We now count the total number of cells without any node among the grid. The code to compute the empty cell test result is directly implemented in this part code. Using a simple bash file, we can execute the simulation 5000 times to plot the result of our test.



We have plotted in blue the result of our 1000 simulations and in orange the theorical limit distribution which is the normal distribution of parameters $\left(\frac{n}{e}, \sqrt{\frac{n(1-2/e)}{e}}\right)$.

We observe that both plots a pretty close. The differences, for example for 37 empty cells, seems to be more due to the low level of simulation than to a real difference in the output. As such, due to the resemblance between the normal distribution and our simulation results, we can say that the empty cell test has a positive result.